RECALL Complex ions AQ-1

Define the term ligand. [1]

RECALL Complex ions AQ-2

Define a co-ordinate (dative covalent) bond. [1]

RECALL Complex ions AQ-3

Define a complex (complex ion). [1]

RECALL Complex ions AQ-4

Define co-ordination number. [1]

RECALL Complex ions AQ-5

Which atom in a water ligand donates the lone pair to the metal ion? [1]

RECALL Complex ions AQ-6

Give the shape and bond angle of $[Cu(H_2O)_6]^{2+}$. [2]

RECALL Complex ions AQ-7

Give the formulas of the four metal-aqua ions in this topic. [1]

RECALL Complex ions AQ-8

What do hexa and aqua mean in “hexaaqua ion”? [1]

RECALL Complex ions AQ-2

1. A shared pair of electrons where **both electrons come from the same atom**

RECALL Complex ions AQ-1

1. An atom, ion or molecule that **donates a pair of electrons** to a central metal ion (forms a co-ordinate bond)

Must say “pair” or “lone pair”. “Donates electrons” alone usually loses the mark.

RECALL Complex ions AQ-4

1. The number of **co-ordinate bonds** to the central metal atom or ion

Not “the number of ligands”: that only works for ligands that form one bond each.

RECALL Complex ions AQ-3

1. A central metal atom or ion **surrounded by ligands** (bonded by co-ordinate bonds)

RECALL Complex ions AQ-6

1. **Octahedral**
2. **90°**

RECALL Complex ions AQ-5

1. The **oxygen** atom

RECALL Complex ions AQ-8

1. Hexa = **six**; aqua = **water** (six water ligands)

RECALL Complex ions AQ-7

1. $[Cu(H_2O)_6]^{2+}$, $[Fe(H_2O)_6]^{2+}$, $[Fe(H_2O)_6]^{3+}$ and $[Al(H_2O)_6]^{3+}$

RECALL Colours AQ-9

Colour of $[Cu(H_2O)_6]^{2+}(aq)$? [1]

RECALL Colours AQ-10

Colour of $[Fe(H_2O)_6]^{2+}(aq)$? [1]

RECALL Colours AQ-11

Colour of $[Fe(H_2O)_6]^{3+}(aq)$? [1]

RECALL Colours AQ-12

Colour of $[Al(H_2O)_6]^{3+}(aq)$? [1]

RECALL Hydroxide precipitates AQ-13

Observation: NaOH(aq) added to $[Cu(H_2O)_6]^{2+}(aq)$. [1]

RECALL Hydroxide precipitates AQ-14

Observation: NaOH(aq) added to $[Fe(H_2O)_6]^{2+}(aq)$. [1]

RECALL Hydroxide precipitates AQ-15

Observation: NaOH(aq) added to $[Fe(H_2O)_6]^{3+}(aq)$. [1]

RECALL Hydroxide precipitates AQ-16

Observation: NaOH(aq) added to $[Al(H_2O)_6]^{3+}(aq)$. [1]

RECALL Colours AQ-10

1. **Green** (pale green) solution

RECALL Colours AQ-9

1. **Blue** solution

RECALL Colours AQ-12

1. **Colourless** solution

Write “colourless”, never “clear”. Clear means see-through, which a blue solution also is.

RECALL Colours AQ-11

1. **Purple/violet**, but usually looks **yellow-brown**

It looks yellow-brown because some is hydrolysed to $[Fe(H_2O)_5(OH)]^{2+}$.

RECALL Hydroxide precipitates AQ-14

1. Green solution → **green precipitate**

It then slowly turns brown on standing in air.

RECALL Hydroxide precipitates AQ-13

1. Blue solution → **blue precipitate**

RECALL Hydroxide precipitates AQ-16

1. Colourless solution → **white precipitate**

RECALL Hydroxide precipitates AQ-15

1. Yellow-brown solution → **brown (orange-brown) precipitate**

RECALL Hydroxide precipitates AQ-17

Formulas of the hydroxide precipitates of a 2+ ion and a 3+ ion. [2]

RECALL Hydroxide precipitates AQ-18

What is the overall charge on a metal hydroxide precipitate such as $[Fe(H_2O)_3(OH)_3]$? [1]

RECALL Hydroxide precipitates AQ-19

Why does the green precipitate from Fe^{2+} turn brown on standing? [1]

RECALL Hydroxide precipitates AQ-20

What role does OH^- (or NH_3) play when it forms a hydroxide precipitate? [1]

RECALL Excess ammonia AQ-21

Which of the four hydroxide precipitates dissolves in excess ammonia? [1]

RECALL Excess ammonia AQ-22

Formula and colour of the complex formed with excess NH_3 and Cu^{2+}. [2]

RECALL Excess ammonia AQ-23

Name the type of reaction when NH_3 replaces H_2O ligands. [1]

RECALL Amphoteric aluminium AQ-24

Define amphoteric. [1]

RECALL Hydroxide precipitates AQ-18

1. **Zero**: it is neutral, which is why it is insoluble

RECALL Hydroxide precipitates AQ-17

1. 2+: **$[M(H_2O)_4(OH)_2]$**
2. 3+: **$[M(H_2O)_3(OH)_3]$**

Charge = number of OH. Waters + OH always add to 6. No overall charge.

RECALL Hydroxide precipitates AQ-20

1. It acts as a **Brønsted–Lowry base**: it accepts H^+ from a water ligand

RECALL Hydroxide precipitates AQ-19

1. It is **oxidised** by oxygen in the air from **Fe(II) to Fe(III)**

RECALL Excess ammonia AQ-22

1. **$[Cu(NH_3)_4(H_2O)_2]^{2+}$**
2. **Deep (dark) blue solution**

Order inside the bracket doesn't matter. "Blue" alone is not enough: say deep or dark blue.

RECALL Excess ammonia AQ-21

1. Only the **copper(II)** one, $[Cu(H_2O)_4(OH)_2]$

RECALL Amphoteric aluminium AQ-24

1. Reacts with (dissolves in) **both acids and bases**

RECALL Excess ammonia AQ-23

1. **Ligand substitution** (ligand exchange)

RECALL Amphoteric aluminium AQ-25

Which metal hydroxide in this topic is amphoteric? [1]

RECALL Amphoteric aluminium AQ-26

Formula and appearance of the product when excess NaOH is added to aluminium hydroxide. [2]

RECALL Carbonate AQ-27

Observation: $Na_2CO_3(aq)$ added to $[Cu(H_2O)_6]^{2+}(aq)$. [1]

RECALL Carbonate AQ-28

Observation: $Na_2CO_3(aq)$ added to $[Fe(H_2O)_6]^{2+}(aq)$. [1]

RECALL Carbonate AQ-29

Observations: $Na_2CO_3(aq)$ added to $[Fe(H_2O)_6]^{3+}(aq)$. [2]

RECALL Carbonate AQ-30

Observations: $Na_2CO_3(aq)$ added to $[Al(H_2O)_6]^{3+}(aq)$. [2]

RECALL Carbonate AQ-31

Name the gas given off when carbonate is added to a 3+ aqua ion. [1]

RECALL Carbonate AQ-32

Which ions form a metal carbonate precipitate with Na_2CO_3, and which form a hydroxide? [1]

RECALL Amphoteric aluminium AQ-26

1. **$[Al(OH)_4]^-$** (or $[Al(H_2O)_2(OH)_4]^-$)
2. White precipitate dissolves → **colourless solution**

RECALL Amphoteric aluminium AQ-25

1. **Aluminium hydroxide**, $[Al(H_2O)_3(OH)_3]$

RECALL Carbonate AQ-28

1. **Green precipitate** ($FeCO_3$), no effervescence

RECALL Carbonate AQ-27

1. **Blue-green precipitate** ($CuCO_3$), no effervescence

RECALL Carbonate AQ-30

1. **White precipitate**
2. **Effervescence** (bubbles of gas)

RECALL Carbonate AQ-29

1. **Brown precipitate**
2. **Effervescence** (bubbles of gas)

RECALL Carbonate AQ-32

1. **2+** ions (Cu^{2+}, Fe^{2+}) → MCO_3; **3+** ions (Fe^{3+}, Al^{3+}) → $M(H_2O)_3(OH)_3 + CO_2$

Trick: three fizzes, two doesn't.

RECALL Carbonate AQ-31

1. **Carbon dioxide**, CO_2

RECALL Chloride ions AQ-33

Formula and colour of the complex formed with concentrated HCl and Cu^{2+}. [2]

RECALL Chloride ions AQ-34

Shape and bond angle of $[CuCl_4]^{2-}$. [2]

RECALL Acidity AQ-35

Which is more acidic: $[Fe(H_2O)_6]^{2+}$ or $[Fe(H_2O)_6]^{3+}$? [1]

RECALL Identify & practical AQ-36

Name one reagent that distinguishes $Fe^{2+}(aq)$ from $Fe^{3+}(aq)$, with both results. [1]

APPLY Hydroxide precipitates AQ-37

Equation: $[Cu(H_2O)_6]^{2+}$ with NaOH(aq). [1]

APPLY Hydroxide precipitates AQ-38

Equation: $[Fe(H_2O)_6]^{2+}$ with NaOH(aq). [1]

APPLY Hydroxide precipitates AQ-39

Equation: $[Fe(H_2O)_6]^{3+}$ with NaOH(aq). [1]

APPLY Hydroxide precipitates AQ-40

Equation: $[Al(H_2O)_6]^{3+}$ with NaOH(aq). [1]

RECALL Chloride ions AQ-34

1. **Tetrahedral**
2. **109.5°**

RECALL Chloride ions AQ-33

1. **$[CuCl_4]^{2-}$**
2. **Yellow** solution (may look green)

RECALL Identify & practical AQ-36

1. NaOH(aq) (or NH_3 or Na_2CO_3): Fe^{2+} **green ppt**; Fe^{3+} **brown ppt**

RECALL Acidity AQ-35

1. **$[Fe(H_2O)_6]^{3+}$** (M^{3+} is always more acidic than M^{2+})

APPLY Hydroxide precipitates AQ-38

1. $[Fe(H_2O)_6]^{2+} + 2OH^- \rightarrow [Fe(H_2O)_4(OH)_2] + 2H_2O$

APPLY Hydroxide precipitates AQ-37

1. $[Cu(H_2O)_6]^{2+} + 2OH^- \rightarrow [Cu(H_2O)_4(OH)_2] + 2H_2O$

State symbols only if asked. Balanced is essential.

APPLY Hydroxide precipitates AQ-40

1. $[Al(H_2O)_6]^{3+} + 3OH^- \rightarrow [Al(H_2O)_3(OH)_3] + 3H_2O$

APPLY Hydroxide precipitates AQ-39

1. $[Fe(H_2O)_6]^{3+} + 3OH^- \rightarrow [Fe(H_2O)_3(OH)_3] + 3H_2O$

APPLY Hydroxide precipitates AQ-41

Equation: $[Cu(H_2O)_6]^{2+}$ with a little $NH_3(aq)$. [1]

APPLY Hydroxide precipitates AQ-42

Equation: $[Fe(H_2O)_6]^{2+}$ with $NH_3(aq)$. [1]

APPLY Hydroxide precipitates AQ-43

Equation: $[Fe(H_2O)_6]^{3+}$ with $NH_3(aq)$. [1]

APPLY Hydroxide precipitates AQ-44

Equation: $[Al(H_2O)_6]^{3+}$ with $NH_3(aq)$. [1]

APPLY Hydroxide precipitates AQ-45

Explain why adding a base to a metal-aqua ion forms a precipitate. [2]

APPLY Hydroxide precipitates not in notes AQ-46

Write an equation for the green Fe(II) precipitate turning brown in air. [2]

APPLY Excess ammonia AQ-47

Equation: excess NH_3 with $[Cu(H_2O)_6]^{2+}$. [1]

APPLY Excess ammonia AQ-48

Equation: excess NH_3 with the copper hydroxide precipitate. [1]

APPLY Hydroxide precipitates AQ-42

1. $[Fe(H_2O)_6]^{2+} + 2NH_3 \rightarrow [Fe(H_2O)_4(OH)_2] + 2NH_4^+$

APPLY Hydroxide precipitates AQ-41

1. $[Cu(H_2O)_6]^{2+} + 2NH_3 \rightarrow [Cu(H_2O)_4(OH)_2] + 2NH_4^+$

NH_3 takes H^+, so the by-product is NH_4^+, not H_2O.

APPLY Hydroxide precipitates AQ-44

1. $[Al(H_2O)_6]^{3+} + 3NH_3 \rightarrow [Al(H_2O)_3(OH)_3] + 3NH_4^+$

APPLY Hydroxide precipitates AQ-43

1. $[Fe(H_2O)_6]^{3+} + 3NH_3 \rightarrow [Fe(H_2O)_3(OH)_3] + 3NH_4^+$

APPLY Hydroxide precipitates not in notes AQ-46

1. $4[Fe(H_2O)_4(OH)_2] + O_2 \rightarrow 4[Fe(H_2O)_3(OH)_3] + 2H_2O$
2. Fe(II) is **oxidised** to Fe(III)

Not in your notes as an equation; the notes only state the change.

APPLY Hydroxide precipitates AQ-45

1. The base **removes H^+** from water ligands (deprotonation), turning them into OH^- ligands
2. Until the complex is **neutral** (no charge), so it is **insoluble**

APPLY Excess ammonia AQ-48

1. $[Cu(H_2O)_4(OH)_2] + 4NH_3 \rightarrow [Cu(NH_3)_4(H_2O)_2]^{2+} + 2OH^- + 2H_2O$

Check the charge: 0 on the left, 2+ and 2(−1) = 0 on the right.

APPLY Excess ammonia AQ-47

1. $[Cu(H_2O)_6]^{2+} + 4NH_3 \rightarrow [Cu(NH_3)_4(H_2O)_2]^{2+} + 4H_2O$

APPLY Excess ammonia not in notes AQ-49

Explain why the co-ordination number stays 6 when NH_3 replaces H_2O ligands. [2]

APPLY Excess ammonia AQ-50

NH_3 can act as a base or as a ligand. Explain both. [2]

APPLY Amphoteric aluminium AQ-51

Equation: aluminium hydroxide acting as a base (with acid). [1]

APPLY Amphoteric aluminium AQ-52

Equation: aluminium hydroxide acting as an acid (with excess OH^-). [1]

APPLY Amphoteric aluminium AQ-53

Equation: $[Al(H_2O)_6]^{3+}$ with excess NaOH "all at once". [1]

APPLY Amphoteric aluminium AQ-54

Explain why aluminium hydroxide dissolves in excess NaOH but not in excess NH_3. [2]

APPLY Carbonate AQ-55

Equation: $[Cu(H_2O)_6]^{2+}$ with $Na_2CO_3(aq)$. [1]

APPLY Carbonate AQ-56

Equation: $[Fe(H_2O)_6]^{2+}$ with $Na_2CO_3(aq)$. [1]

APPLY Excess ammonia AQ-50

1. Base: accepts an **H^+** from a water ligand (forms NH_4^+)
2. Ligand: donates its **lone pair on N** to the metal ion (co-ordinate bond)

APPLY Excess ammonia not in notes AQ-49

1. NH_3 and H_2O are **similar in size**
2. Both are **uncharged** (neutral) ligands

Your notes cover the substitution; this explanation is from the AQA spec (3.2.5).

APPLY Amphoteric aluminium AQ-52

1. $[Al(H_2O)_3(OH)_3] + OH^- \rightarrow [Al(OH)_4]^- + 3H_2O$

Also accepted: $\rightarrow [Al(H_2O)_2(OH)_4]^- + H_2O$ (only 1 water then).

APPLY Amphoteric aluminium AQ-51

1. $[Al(H_2O)_3(OH)_3] + 3H^+ \rightarrow [Al(H_2O)_6]^{3+}$

APPLY Amphoteric aluminium AQ-54

1. NaOH provides a **high concentration of OH^-**, which removes a further H^+ to form soluble $[Al(OH)_4]^-$
2. NH_3 is a **weak base**: the $[OH^-]$ is too low to do this

APPLY Amphoteric aluminium AQ-53

1. $[Al(H_2O)_6]^{3+} + 4OH^- \rightarrow [Al(OH)_4]^- + 6H_2O$

APPLY Carbonate AQ-56

1. $[Fe(H_2O)_6]^{2+} + CO_3^{2-} \rightarrow FeCO_3 + 6H_2O$

APPLY Carbonate AQ-55

1. $[Cu(H_2O)_6]^{2+} + CO_3^{2-} \rightarrow CuCO_3 + 6H_2O$

APPLY Carbonate AQ-57

Equation: $[Fe(H_2O)_6]^{3+}$ with $Na_2CO_3(aq)$. [1]

APPLY Carbonate AQ-58

Equation: $[Al(H_2O)_6]^{3+}$ with $Na_2CO_3(aq)$. [1]

APPLY Chloride ions AQ-59

Equation: $[Cu(H_2O)_6]^{2+}$ with concentrated HCl. [1]

APPLY Chloride ions AQ-60

Explain why the co-ordination number changes from 6 to 4 when Cl^- replaces H_2O. [2]

APPLY Chloride ions not in notes AQ-61

How could you reverse the formation of $[CuCl_4]^{2-}$? [1]

APPLY Acidity AQ-62

Equation: $[Fe(H_2O)_6]^{3+}$ acting as an acid in water. [1]

APPLY Acidity AQ-63

Equation: $[Al(H_2O)_6]^{3+}$ acting as an acid in water. [1]

APPLY Acidity AQ-64

Equation: $[Fe(H_2O)_6]^{2+}$ acting as an acid in water. [1]

APPLY Carbonate AQ-58

1. $2[Al(H_2O)_6]^{3+} + 3CO_3^{2-} \rightarrow 2[Al(H_2O)_3(OH)_3] + 3CO_2 + 3H_2O$

APPLY Carbonate AQ-57

1. $2[Fe(H_2O)_6]^{3+} + 3CO_3^{2-} \rightarrow 2[Fe(H_2O)_3(OH)_3] + 3CO_2 + 3H_2O$

Watch the charge on Fe: 3+. 2 × 3+ balances 3 × 2−.

APPLY Chloride ions AQ-60

1. Cl^- ions are **larger** than H_2O molecules
2. So **fewer fit** around the metal ion (only 4)

APPLY Chloride ions AQ-59

1. $[Cu(H_2O)_6]^{2+} + 4Cl^- \rightleftharpoons [CuCl_4]^{2-} + 6H_2O$

APPLY Acidity AQ-62

1. $[Fe(H_2O)_6]^{3+} + H_2O \rightleftharpoons [Fe(H_2O)_5(OH)]^{2+} + H_3O^+$

Also accepted: $[Fe(H_2O)_6]^{3+} \rightleftharpoons [Fe(H_2O)_5(OH)]^{2+} + H^+$

APPLY Chloride ions not in notes AQ-61

1. **Add water** (dilute it): the equilibrium shifts left, back to blue $[Cu(H_2O)_6]^{2+}$

Not in your notes, but follows from the ⇌ in the equation.

APPLY Acidity AQ-64

1. $[Fe(H_2O)_6]^{2+} + H_2O \rightleftharpoons [Fe(H_2O)_5(OH)]^{+} + H_3O^+$

Happens to a much smaller extent than for 3+ ions.

APPLY Acidity AQ-63

1. $[Al(H_2O)_6]^{3+} + H_2O \rightleftharpoons [Al(H_2O)_5(OH)]^{2+} + H_3O^+$

APPLY Acidity AQ-65

Explain why a solution of iron(III) chloride is acidic. [2]

APPLY Colours AQ-66

Why does $[Fe(H_2O)_6]^{3+}(aq)$ usually look yellow-brown, not purple? [1]

APPLY Complex ions AQ-67

Give the charge of each: $[Fe(H_2O)_5(OH)]^x$ and $[Al(OH)_4]^y$. [2]

APPLY Identify & practical AQ-68

A solution gives a white ppt with NaOH(aq) that dissolves in excess. Identify the ion and the final complex. [2]

APPLY Identify & practical AQ-69

A blue solution gives a blue ppt with a little NH_3 and a deep blue solution with excess. Identify the ion and final complex. [2]

EXAM Acidity AQ-70

Explain, in terms of charge/size ratio, why $[Al(H_2O)_6]^{3+}$ is more acidic than $[Fe(H_2O)_6]^{2+}$. [3]

EXAM Carbonate AQ-71

Describe and explain what you see when $Na_2CO_3(aq)$ is added separately to $Fe^{2+}(aq)$ and $Fe^{3+}(aq)$. [4]

APPLY Colours AQ-66

1. Some is hydrolysed to **$[Fe(H_2O)_5(OH)]^{2+}$**, which is yellow-brown

APPLY Acidity AQ-65

1. Water ligands lose H^+ (the aqua ion is hydrolysed)
2. $[Fe(H_2O)_6]^{3+} + H_2O \rightleftharpoons [Fe(H_2O)_5(OH)]^{2+} + H_3O^+$, so **$H_3O^+$** ions are produced

APPLY Identify & practical AQ-68

1. **Al^{3+}**
2. **$[Al(OH)_4]^-$** (colourless solution)

APPLY Complex ions AQ-67

1. x = **2+** (3+ and one OH^-)
2. y = **1−** (3+ and four OH^-)

Charge = metal charge + (−1 for each OH^- or Cl^-; 0 for H_2O or NH_3).

EXAM Acidity AQ-70

1. Al^{3+} is **smaller** and has a **higher charge**, so a higher charge/size ratio (charge density)
2. So it is **more polarising** and **weakens the O–H bonds** in the water ligands
3. So **H^+ is released more easily**

All three links needed for 3 marks.

APPLY Identify & practical AQ-69

1. **Cu^{2+}** ($[Cu(H_2O)_6]^{2+}$)
2. **$[Cu(NH_3)_4(H_2O)_2]^{2+}$**

EXAM Carbonate AQ-71

1. Fe^{2+}: **green ppt** of $FeCO_3$, no effervescence
2. Fe^{3+}: **brown ppt** and **effervescence**
3. $[Fe(H_2O)_6]^{3+}$ is **more acidic** (higher charge/size): releases H^+/H_3O^+
4. H_3O^+ reacts with CO_3^{2-} to give **CO_2**; the 2+ ion is not acidic enough

EXAM Amphoteric aluminium AQ-72

Use equations to show that aluminium hydroxide is amphoteric. [3]

EXAM Chloride ions AQ-74

Concentrated HCl is added to $[Cu(H_2O)_6]^{2+}(aq)$. Give the equation, the colour change, the new co-ordination number and shape. [4]

EXAM Excess ammonia AQ-73

Describe what you see when $NH_3(aq)$ is added dropwise then in excess to $Cu^{2+}(aq)$. Give equations. [4]

EXAM Chloride ions AQ-75

Compare the substitution of H_2O by NH_3 and by Cl^- in $[Cu(H_2O)_6]^{2+}$, explaining any change in co-ordination number. [4]

EXAM Hydroxide precipitates AQ-76

$Fe^{2+}(aq)$ is treated with NaOH(aq) and the mixture is left in air. Describe all observations and explain the final change. [3]

EXAM Complex ions AQ-80

Describe the bonding in $[Fe(H_2O)_6]^{3+}$ and state its shape. [3]

EXAM Chloride ions AQ-74

1. $[Cu(H_2O)_6]^{2+} + 4Cl^- \rightleftharpoons [CuCl_4]^{2-} + 6H_2O$
2. **Blue → yellow** (green) solution
3. Co-ordination number **4**
4. **Tetrahedral** (Cl^- larger, so fewer fit)

EXAM Amphoteric aluminium AQ-72

1. States amphoteric = reacts with acids and bases
2. $[Al(H_2O)_3(OH)_3] + 3H^+ \rightarrow [Al(H_2O)_6]^{3+}$
3. $[Al(H_2O)_3(OH)_3] + OH^- \rightarrow [Al(OH)_4]^- + 3H_2O$

EXAM Excess ammonia AQ-73

1. **Blue ppt** forms
2. $[Cu(H_2O)_6]^{2+} + 2NH_3 \rightarrow [Cu(H_2O)_4(OH)_2] + 2NH_4^+$
3. Ppt dissolves to a **deep blue solution**
4. $[Cu(H_2O)_4(OH)_2] + 4NH_3 \rightarrow [Cu(NH_3)_4(H_2O)_2]^{2+} + 2OH^- + 2H_2O$

EXAM Chloride ions AQ-75

1. NH_3: co-ordination number **stays 6**, octahedral
2. NH_3 and H_2O are **similar size** and **uncharged**
3. Cl^-: co-ordination number **6 → 4**, tetrahedral
4. Cl^- is **larger** than H_2O, so fewer fit

The explanation for NH_3 is from the AQA spec (3.2.5).

EXAM Complex ions AQ-80

1. Each H_2O **donates a lone pair** from oxygen to Fe^{3+}
2. Forming 6 **co-ordinate (dative covalent) bonds**
3. **Octahedral**, 90°

EXAM Hydroxide precipitates AQ-76

1. **Green ppt** forms ($[Fe(H_2O)_4(OH)_2]$)
2. **No change in excess** NaOH
3. On standing it turns **brown**: Fe(II) oxidised to Fe(III) by **oxygen in air**

EXAM Identify & practical AQ-77

Solution X gives a white ppt with NaOH(aq) that dissolves in excess, and a white ppt with effervescence with Na_2CO_3(aq). Identify X and write an equation for each reaction. [4]

EXAM Identify & practical not in notes AQ-78

Required practical 11. You have four unlabelled solutions: Cu^{2+}, Fe^{2+}, Fe^{3+} and Al^{3+}. Describe how to identify each using test-tube reactions. [6, levels] [6]

EXAM Excess ammonia AQ-79

Describe what you see when NH_3(aq) is added dropwise then in excess to separate solutions of Fe^{2+}, Cu^{2+}, Fe^{3+} and Al^{3+}. Explain the differences. [6, levels] [6]

EXAM Identify & practical AQ-81

Copper metal is extracted from Cu^{2+}(aq) using scrap iron. Give the equation and the colours of the initial and final solutions. [3]

EXAM Acidity AQ-83

Suggest why $[Fe(H_2O)_6]^{3+}$ and $[Al(H_2O)_6]^{3+}$ react with carbonate to give CO_2 but $[Cu(H_2O)_6]^{2+}$ does not. [2]

EXAM Identify & practical AQ-77

1. X contains **Al^{3+}** ($[Al(H_2O)_6]^{3+}$)
2. $[Al(H_2O)_6]^{3+} + 3OH^- \rightarrow [Al(H_2O)_3(OH)_3] + 3H_2O$
3. $[Al(H_2O)_3(OH)_3] + OH^- \rightarrow [Al(OH)_4]^- + 3H_2O$
4. $2[Al(H_2O)_6]^{3+} + 3CO_3^{2-} \rightarrow 2[Al(H_2O)_3(OH)_3] + 3CO_2 + 3H_2O$

EXAM Identify & practical not in notes AQ-78

1. Small sample of each in a test tube; add **NaOH(aq) dropwise**, then **in excess**
2. Cu^{2+} **blue** ppt; Fe^{2+} **green** ppt (browns in air); Fe^{3+} **brown** ppt
3. Al^{3+} **white** ppt that **dissolves in excess** NaOH
4. Confirm with NH_3: only Cu^{2+} ppt dissolves to **deep blue**
5. Or Na_2CO_3: Fe^{3+} and Al^{3+} **effervesce**; Fe^{2+}/Cu^{2+} do not
6. Initial colours also help: blue, green, yellow-brown, colourless

Levels marking: 5–6 marks needs a clear method plus correct results for all four ions, including the excess step.

EXAM Excess ammonia AQ-79

1. Fe^{2+}: green ppt, no change in excess
2. Cu^{2+}: blue ppt, dissolves in excess → **deep blue solution**
3. Fe^{3+}: brown ppt, no change in excess
4. Al^{3+}: white ppt, **no change** in excess
5. Ppt forms because NH_3 acts as a base, removing H^+ until the complex is neutral
6. Only Cu^{2+} undergoes **ligand substitution** with NH_3; NH_3 too weak a base to dissolve $Al(OH)_3$

For full marks give a correct equation too, e.g. $[Fe(H_2O)_6]^{3+} + 3NH_3 \rightarrow [Fe(H_2O)_3(OH)_3] + 3NH_4^+$.

EXAM Acidity AQ-83

1. 3+ ions have a higher charge/size ratio so their solutions contain more **H_3O^+** (more acidic)
2. Enough H_3O^+ to react with CO_3^{2-} to form **CO_2**; 2+ solutions are not acidic enough, so MCO_3 precipitates

EXAM Identify & practical AQ-81

1. $Fe + Cu^{2+} \rightarrow Fe^{2+} + Cu$
2. Initial solution **blue**
3. Final solution **(pale) green**

From the June 2010 question in your notes.

From $[Cu(H_2O)_6]^{2+}(aq)$, identify W (conc. HCl), X (dilute NH_3), Y (excess NH_3) and Z (Na_2CO_3), with appearances. [6]

1. W = $[CuCl_4]^{2-}$, yellow solution
2. X = $[Cu(H_2O)_4(OH)_2]$, blue ppt
3. Y = $[Cu(NH_3)_4(H_2O)_2]^{2+}$, deep blue solution
4. Z = $CuCO_3$, blue-green ppt

Based on June 2010 Q5 in your notes; each part there also needs an equation.